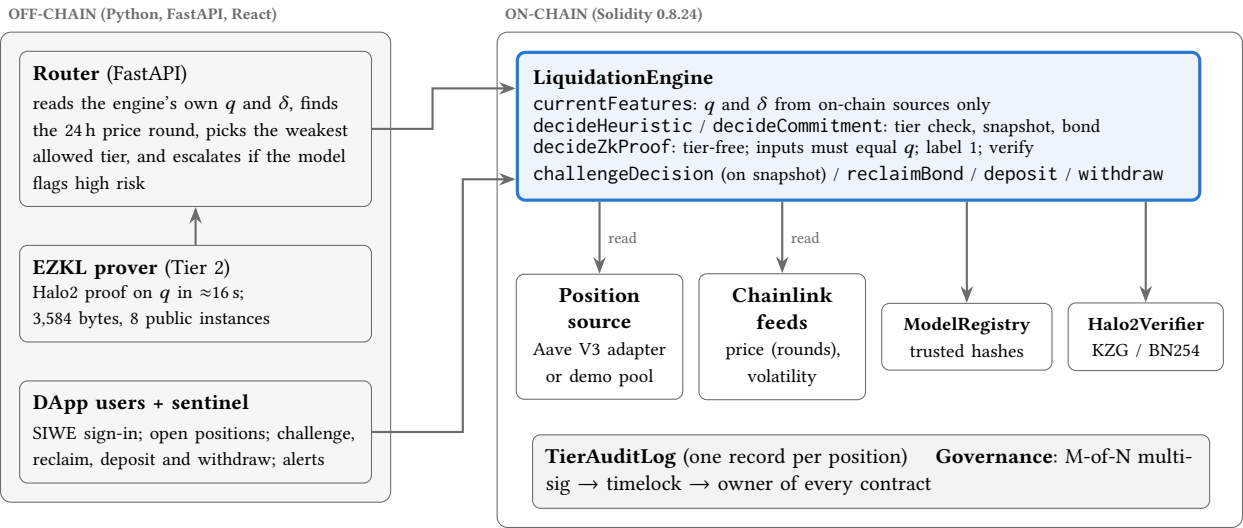

OFF-CHAIN (Python, FastAPI, React)
ON-CHAIN (Solidity 0.8.24)
Router (FastAPI)
reads the engine's own q and δ, finds the 24 h price round, picks the weakest allowed tier, and escalates if the model flags high risk
EZKL prover (Tier 2)
Halo2 proof on q in ≈16 s; 3,584 bytes, 8 public instances
DApp users + sentinel
SIWE sign-in; open positions; challenge, reclaim, deposit and withdraw; alerts
LiquidationEngine
currentFeatures: q and δ from on-chain sources only
decideHeuristic / decideCommitment: tier check, snapshot, bond
decideZkProof: tier-free; inputs must equal q; label 1; verify
challengeDecision (on snapshot) / reclaimBond / deposit / withdraw
read
read
Position source
Aave V3 adapter or demo pool
Chainlink feeds
price (rounds), volatility
ModelRegistry
trusted hashes
Halo2Verifier
KZG / BN254
TierAuditLog (one record per position)
Governance: M-of-N multi-sig → timelock → owner of every contract

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.